

CLINISCOPE

Healthcare Record Contradiction Detector

Find the conflict. Trace the evidence. Keep humans in control.

TEAM [Add official team name]

Team leader: [Add team leader]

Team members: [Add member names]

Problem Statement

PS-11R3 · Healthcare Record Contradiction Detector. The framing below is our own interpretation of the problem.

01 THE PROBLEM

Patient information is spread across discharge summaries, intake forms, medication lists and lab reports. Allergies, doses, diagnoses and lab values can differ between records, and finding those differences by hand is slow.

02 TARGET USERS (INTENDED)

- **Clinicians** reviewing patient records
- **Medical-records / HIM teams**
- **Clinical quality & patient-safety** reviewers
- **Authorized staff** reconciling records

03 CURRENT CHALLENGES

- **Fragmented:** related statements sit in separate documents
- **Context-sensitive:** dates, negation and dose changes blur conflicts
- **Hard to trace:** each discrepancy needs its original source

04 THE GAP / IMPACT

A warning alone is not enough. Reviewers need the original evidence, the context (dates, history) to judge it, and a controlled way to record their final decision.

ONE-LINE PROBLEM: How might we help healthcare reviewers identify and investigate inconsistencies across medical documents, with clear source evidence and human oversight?

Proposed Solution

An evidence-first review prototype: documents in, traceable findings out, decisions made by people.

SOLUTION OVERVIEW

CLINISCOPE extracts text from PDF, TXT and DOCX records, identifies allergy, medication, diagnosis, lab, procedure and smoking statements with deterministic rules, compares them across the documents of one case, and flags potential discrepancies with verified source quotations for human review.

UNIQUE VALUE / USP

Every finding is traceable: it is created only if each quote is found again at its exact source offsets. Dated or documented changes are separated from true conflicts, and the final decision stays with the reviewer.

01 INGEST DOCUMENTS

PDF (text layer), TXT and DOCX are validated, de-duplicated and extracted in the browser.



02 DETECT WITH EVIDENCE

Rules compare statements and label each difference by type, backed by quoted evidence.



03 REVIEW & AUDIT

Confirm, resolve or dismiss with a required reason; every action lands in an audit log.

EXPECTED BENEFIT: Potential inconsistencies become easier to locate and investigate, while evidence traceability and human judgment are preserved.

Technical Approach & Architecture

The whole pipeline runs in the reviewer's browser: there is no backend server, no AI/LLM and no OCR in the current build.



SYSTEM COMPONENTS

- 1 Extraction:** pdf.js, mammoth, UTF-8 text with offsets
- 2 Statements:** negation, hedging, history, dose changes
- 3 Detection:** per-case comparison, units & dates
- 4 Evidence:** quotes re-verified; View in source
- 5 Review:** state machine, notes, audit log

DATA FLOW & VALIDATION

- **Uploads:** type, magic-byte and size checks; dedupe
- **Findings:** only if every quote matches its source
- **Dates:** specimen dates and dose changes reclassify
- **Scanned PDFs:** flagged; every decision needs a reason

ARCHITECTURE NOTE: Extraction, statement rules, detection and review are separate TypeScript modules with unit tests, so each stage can be tested and replaced independently.

Technology Stack

Verified against package.json and the CI workflow. No Python, server framework, cloud database, OCR or LLM provider is used.

FRONTEND

React 18 + TypeScript, built with Vite and styled with Tailwind CSS.

BACKEND: NONE

A TypeScript service layer in the browser runs upload, analysis and review.

STORAGE

IndexedDB via Dexie: cases, documents, original files, findings and audit events.

CORE LOGIC (RULES)

Deterministic TypeScript rules: vocabulary, negation, units and date handling.

DOCUMENT PROCESSING

pdfjs-dist reads PDF text layers, mammoth reads DOCX; TXT read as UTF-8.

TESTING & DEPLOYMENT

Vitest + Playwright in GitHub Actions; static build hosted on GitHub Pages.

WHY THIS STACK? A lightweight, testable static web app: documents are processed and stored in the reviewer's own browser, so no server or credentials are needed. A backend, OCR or AI service would be added only when the implementation requires it.

Feasibility & Viability

A working, tested prototype today, with a staged and cautious path toward real-world evaluation.

TECHNICAL FEASIBILITY

- **Tools & skills:** React, TypeScript, rule design, pdf.js / mammoth, Vitest and Playwright, all in use today.
- **Data & infrastructure:** synthetic demo records and browser storage only; no hospital systems or real patient data.
- **Status & constraints:** deployed prototype; 34 unit + 8 end-to-end tests pass. Limited vocabulary, no OCR, no login.
- **Risk → mitigation:** mis-extraction and context false positives → offset-verified quotes, uncertainty labels, human review, regression tests.

BUSINESS / OPERATIONAL VIABILITY

- **Users & value:** authorized reviewers who reconcile records and need evidence, not just alerts.
- **Adoption path:** synthetic prototype → representative test set → clinical reviewer feedback → security & privacy review → approved limited pilot.
- **Cost drivers:** hosting (static today), document processing, future storage or AI APIs, testing and maintenance.
- **Possible model (future):** institutional licensing or record-system integration via pilot partners. No partners or customers today.

KEY TAKEAWAY: CLINISCOPE builds on a working software prototype and offers a practical path toward evaluated, evidence-based record review, subject to further validation and secure deployment.

Sustainability, Scalability & Impact

Scalability items and roadmap stages 02–04 are planned, not built; impact is a goal to be measured in evaluation.

LONG-TERM SUSTAINABILITY

- Modular engine in src/lib
- Regression tests run in CI
- Versioned DB schema
- Unsupported files flagged openly
- Synthetic, governed test data

SCALABILITY (PLANNED)

- OCR for scanned records
- More formats and vocabulary
- Evidence-checked AI assistance
- Secure backend, login, multi-user
- Record-system integration

EXPECTED IMPACT (GOALS)

- Aims to cut manual searching
- More traceable investigations
- Clearer view of source evidence
- Structured review records
- Better awareness of uncertainty

FUTURE ROADMAP



01 Prototype (now)

Synthetic records: ingestion, evidence, review



02 Pilot testing

Measure extraction, accuracy and usability



03 Improve

OCR, context, evidence checks, security



04 Deploy & scale

Only after accuracy, privacy and approvals

SUCCESS METRICS (to be measured): detection precision & recall on a labelled set • % of findings with verifiable evidence • false-positive rate • reviewer agreement • median time per finding • extraction success rate

Conclusion

CLINISCOPE helps reviewers identify potential inconsistencies across healthcare documents, trace every finding to verified source evidence, and record decisions with human oversight.

Find the conflict. Trace the evidence. Keep humans in control.

THANK YOU

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Live demo: anirudhleetcode-max.github.io/cliniscope-healthcare-contradiction-detector

Code: github.com/anirudhleetcode-max/cliniscope-healthcare-contradiction-detector

Contact: [Add team email]



SCAN FOR LIVE DEMO

CLINISCOPE